

# When Standard Methods Succeed

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**when correlation *is* causation**

When you have no confounders and there is a linear relationship between the exposure and the outcome, that *correlation is a causal relationship*



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# **randomized controlled trials**

## **A/B testing**

**Even in these cases, using the methods  
you will learn here can help!**

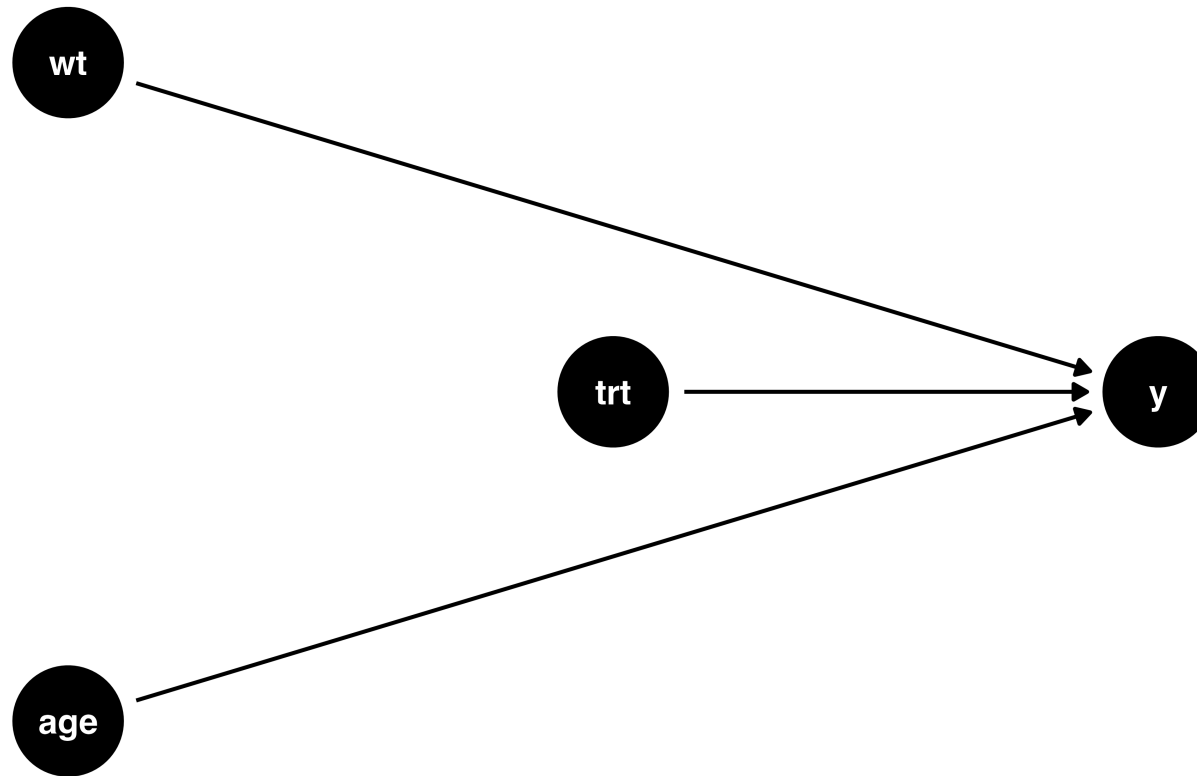


- 1 Adjusting for baseline covariates can make an estimate *more efficient*
- 2 Propensity score weighting is *more efficient* than direct adjustment
- 3 Sometimes we are *more comfortable with the functional form of the propensity score* (predicting exposure) than the outcome model

# Example

- simulated data (100 observations)
- Treatment is randomly assigned
- There are two baseline covariates: age and weight

# Example



- True average treatment effect: 1

## Unadjusted model

```
1 lm(y ~ treatment, data = data)
```

| Characteristic | Beta | SE    | 95% CI     | p-value |
|----------------|------|-------|------------|---------|
| treatment      | 0.93 | 0.803 | -0.66, 2.5 | 0.2     |

Abbreviations: CI = Confidence Interval, SE = Standard Error

## Adjusted model

```
1 lm(y ~ treatment + weight + age, data = data)
```

| Characteristic | Beta | SE    | 95% CI     | p-value |
|----------------|------|-------|------------|---------|
| treatment      | 1.0  | 0.204 | 0.59, 1.4  | <0.001  |
| weight         | 0.34 | 0.106 | 0.13, 0.55 | 0.002   |
| age            | 0.20 | 0.005 | 0.19, 0.22 | <0.001  |

Abbreviations: CI = Confidence Interval, SE = Standard Error

## Propensity score adjusted model

| Characteristic | Beta | SE    | 95% CI   | p-value |
|----------------|------|-------|----------|---------|
| treatment      | 1    | 0.202 | 0.6, 1.4 | <0.001  |

# Example

- simulated data (10,000 observations)
- Treatment is randomly assigned
- There are two baseline covariates: age and weight

## Unadjusted model

```
1 lm(y ~ treatment, data = data)
```

| Characteristic | Beta | SE    | 95% CI    | p-value |
|----------------|------|-------|-----------|---------|
| treatment      | 0.96 | 0.083 | 0.80, 1.1 | <0.001  |

Abbreviations: CI = Confidence Interval, SE = Standard Error

## Adjusted model

```
1 lm(y ~ treatment + weight + age, data = data)
```

| Characteristic | Beta | SE    | 95% CI     | p-value |
|----------------|------|-------|------------|---------|
| treatment      | 1.0  | 0.020 | 0.98, 1.1  | <0.001  |
| weight         | 0.20 | 0.010 | 0.18, 0.22 | <0.001  |
| age            | 0.20 | 0.000 | 0.20, 0.20 | <0.001  |

Abbreviations: CI = Confidence Interval, SE = Standard Error

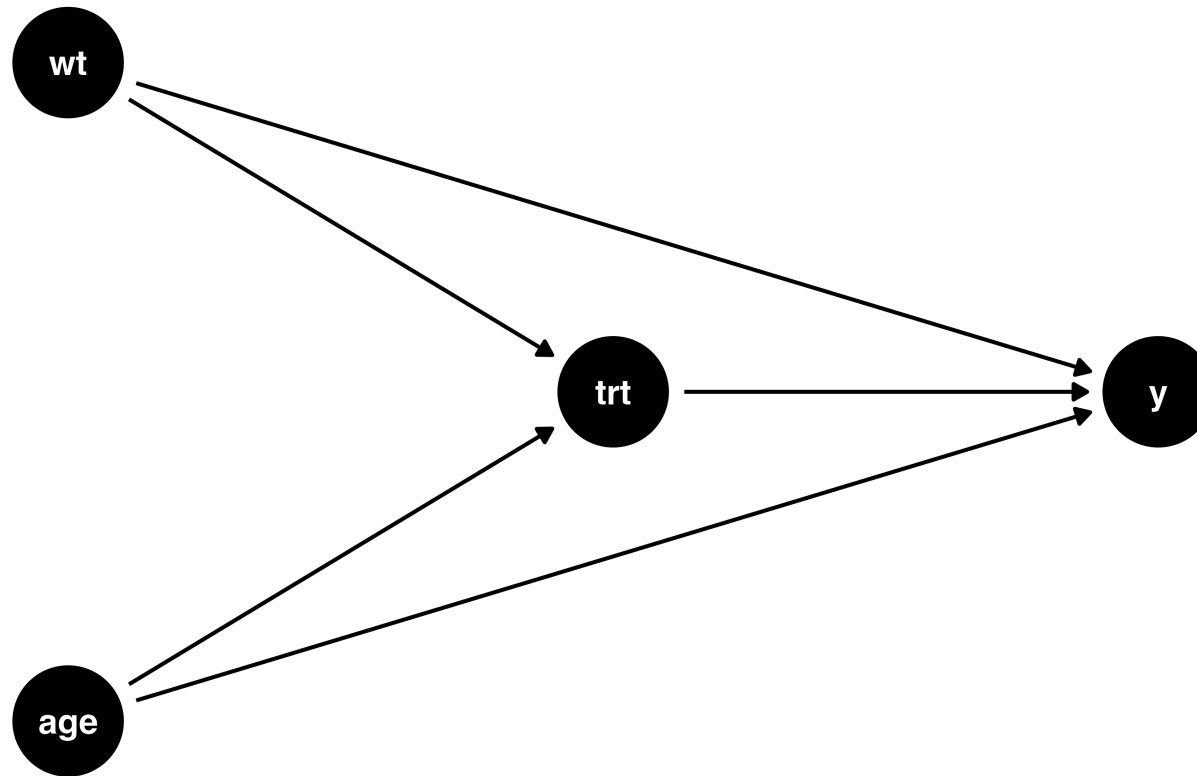
## Propensity score adjusted model

| Characteristic | Beta | SE   | 95% CI | p-value |
|----------------|------|------|--------|---------|
| treatment      | 1    | 0.02 | 1, 1.1 | <0.001  |

# Example

- simulated data (10,000 observations)
- Treatment is **not** randomly assigned
- There are **two baseline confounders**: age and weight
- The treatment effect is **homogeneous**

# Example



- True average treatment effect: 1



## Unadjusted model

```
1 lm(y ~ treatment, data = data)
```

| Characteristic | Beta | SE    | 95% CI   | p-value |
|----------------|------|-------|----------|---------|
| treatment      | 1.8  | 0.085 | 1.7, 2.0 | <0.001  |

Abbreviations: CI = Confidence Interval, SE = Standard Error

## Adjusted model

```
1 lm(y ~ treatment + weight + age, data = data)
```

| Characteristic | Beta | SE    | 95% CI     | p-value |
|----------------|------|-------|------------|---------|
| treatment      | 0.98 | 0.021 | 0.94, 1.0  | <0.001  |
| weight         | 0.20 | 0.010 | 0.18, 0.22 | <0.001  |
| age            | 0.20 | 0.000 | 0.20, 0.20 | <0.001  |

Abbreviations: CI = Confidence Interval, SE = Standard Error

## Propensity score adjusted model

| Characteristic | Beta | SE    | 95% CI | p-value |
|----------------|------|-------|--------|---------|
| treatment      | 1    | 0.022 | 0.9, 1 | <0.001  |

***time-varying* confounding**